No of additional sheets: 1

LOURDES MATHA COLLEGE OF SCIENCE & TECHNOLOGY
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ANSWER BOOK

Name: Adwaith.A.M Class No: 05 Branch: MCA

Subject: Deep learning Semester: S3

Mark Tabulation

Mark Divisions	Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	Q.7	Q.8	Q.9	Q.10	Q.11	Q.12	Q.13	Q.14	Q.15
	1	½		2		1	1	½							
Total															
Grand Total													6		

1, Sigmoid activation functions are functions that trigger or activate ~~nodes~~ needs in a neural network. sigmoid ~~activation~~ ~~is~~ ~~neuron~~ function is used as an activation function. ~~most~~ ~~to~~ ~~used~~ ~~as~~ ~~activation~~ ~~function~~ when the activation function for a neuron is a sigmoid function it is guaranteed that the output of this unit will always be between 0 and 1. Also as the sigmoid is a non-linear function, the output of this unit would be a non-linear function of the weighted sum of inputs. such a neuron which employs a sigmoid function ~~that~~ as an

→ steepness parameter

5. activation function is termed as a sigmoid unit.

2. Gradient descent is known as one of the most commonly used optimisation algorithm, to train machine learning models by means of minimizing errors between actual and unexpected results. Further gradient descent is also used to train neural networks. In mathematical terminology optimisation algorithm refers to the task of minimising an objective function $f(x)$ parametrised by x. similarly in machine learning optimisation is the task of minimising the cost function parametrised by the model parameters The main objective of the gradient descent is convex function using iteration of parameters update:

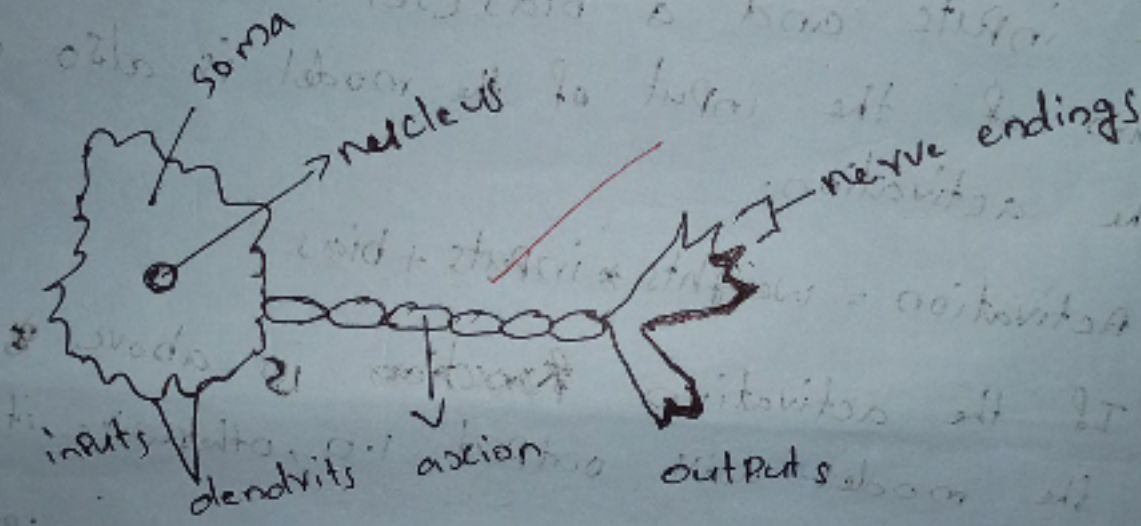
4, The Perceptron algorithm is a two class (binary) classification machine learning algorithm. It is a type of neural network model perhaps the simplest type of neural network model. It consists of a single node or neuron that takes a row of data as input and predicts a class label. This is achieved by calculating the weighted sum of the inputs and a bias (set to 1). The weighted sum of the input of the model is also called the activation.

Activation = weights * inputs + bias

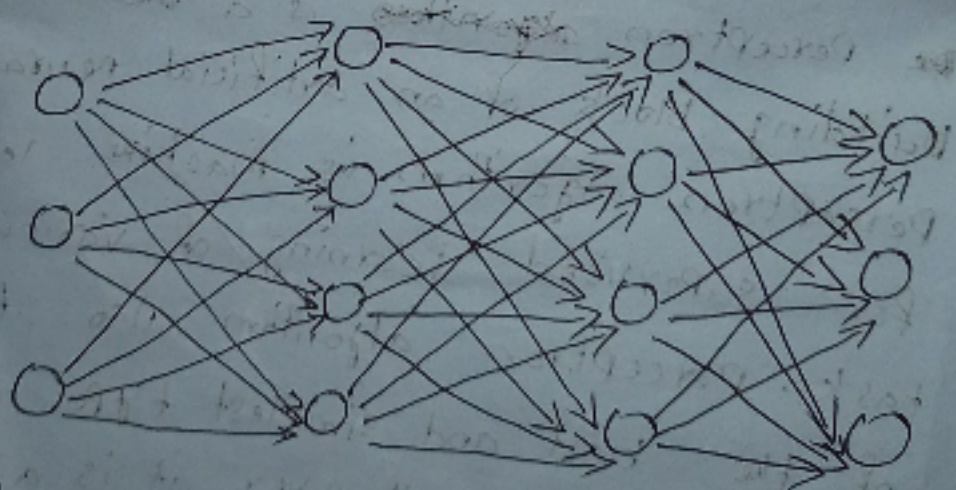
If the activation is above 0.0, the model will output 1.0, otherwise it will output 0.0.

Perceptron is a Building block of an artificial neural network. Perceptron algorithm is machine learning algorithm for supervised learning of various classification tasks. Perceptron algorithm also treated as one of the best and simplest types of artificial neural networks. However it is a supervised algorithm of binary classification.

6, Biological neuron models, also known as neuron models, The mathematical description of the properties of certain cells in the nerve system that generates short electrical potentials across their cells, membrane roughly are millisecond in duration called action potential or spikies.

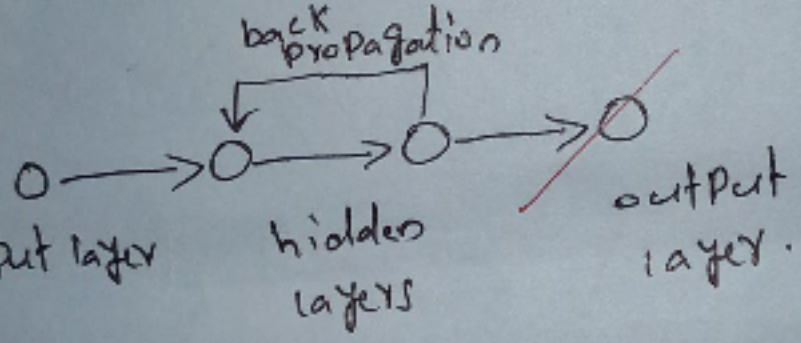


7,



input layer Hidden layers output layer

back Propagation is the essence of neural network training. it is the method of fine tuning the weights of a neural network based on the errors rate obtained in the previous epoch. Back Propagation in neural network as short form for backward Propagation of errors.



8, Batch normalisation is the ~~Process of the~~ Process of normalising of a group of data in training of neural network. overfitting occurs when the data exceeds the size of the batch.